MILITARY SPECIFICATION

PAPER, ELECTROCARDIOGRAPH RECORDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of single channel, heat-pressure sensitive recording paper.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up, Paperboard.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

T404	Tensile Breaking Strength of Paper and Paperboard.
T410	Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T411	Thickness (caliper) of Paper and Paperboard.
T452	Brightness of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 1 Dunwoody Park, Atlanta, GA 30341.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection (see 4.1.2.1 and 6.2).

3.2 Materials.

3.2.1 Coated paper. The coated paper shall be a heat-sensitive and water-resistant type, with the following properties:

Thickness.....0.0031 to 0.0037 (inch
Basis weight (500, 20 by 24 inch sheets).....20 to 28 (pounds)
Tensile strength (minimum)
Machine direction.....4.5 (Kilograms per 15 millimetres width)
Cross machine direction.....2.0 (Kilograms per 15 millimetres width)
Brightness (unprinted).....62 to 74 percent
Water resistance.....No permanent discoloration shall result from test specified.

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3.3 Construction. The electrocardiograph (ECG) recording paper shall be in accordance with figure 1. The grid pattern shall extend the full length of the paper. The grid lines shall not wander and shall be clearly printed in a dark gray or black color of such density that ECG traces can be distinguished from the grid background and can be measured by counting grid lines. The finished paper shall be spooled, heat-pressure sensitive side in, on a core with an inside diameter of 0.627 ± 0.007 inches (15.92 ± 0.18 mm). The core shall not distort or change shape during handling or shipping. The maximum outside diameter of the roll of paper shall be 3.032 inches (77.01 mm). Roll ends shall be smooth straight and parallel.

3.4 Performance. The ECG paper specified shall conform to the performance parameters listed below when used in conjuction with at least one of the ECG machines listed in 6.5.

3.4.1 High speed writing. When tested as specified in 4.4.6, the resultant traces shall be visible.

3.4.2 Amplitude doubling. When the test specified in 4.4.7 is performed, the maximum deviation (of the average stylus deflection) from doubling shall be 0.5 mm.

3.4.3 Buildup of paper coating and degradation of traces. There shall be no buildup of the paper coating on the ECG stylus, and the amplitude, shape and baseline width of the trace shall not change due to buildup of coating when the traces from the start and finish of the test specified in 4.4.8 are compared.

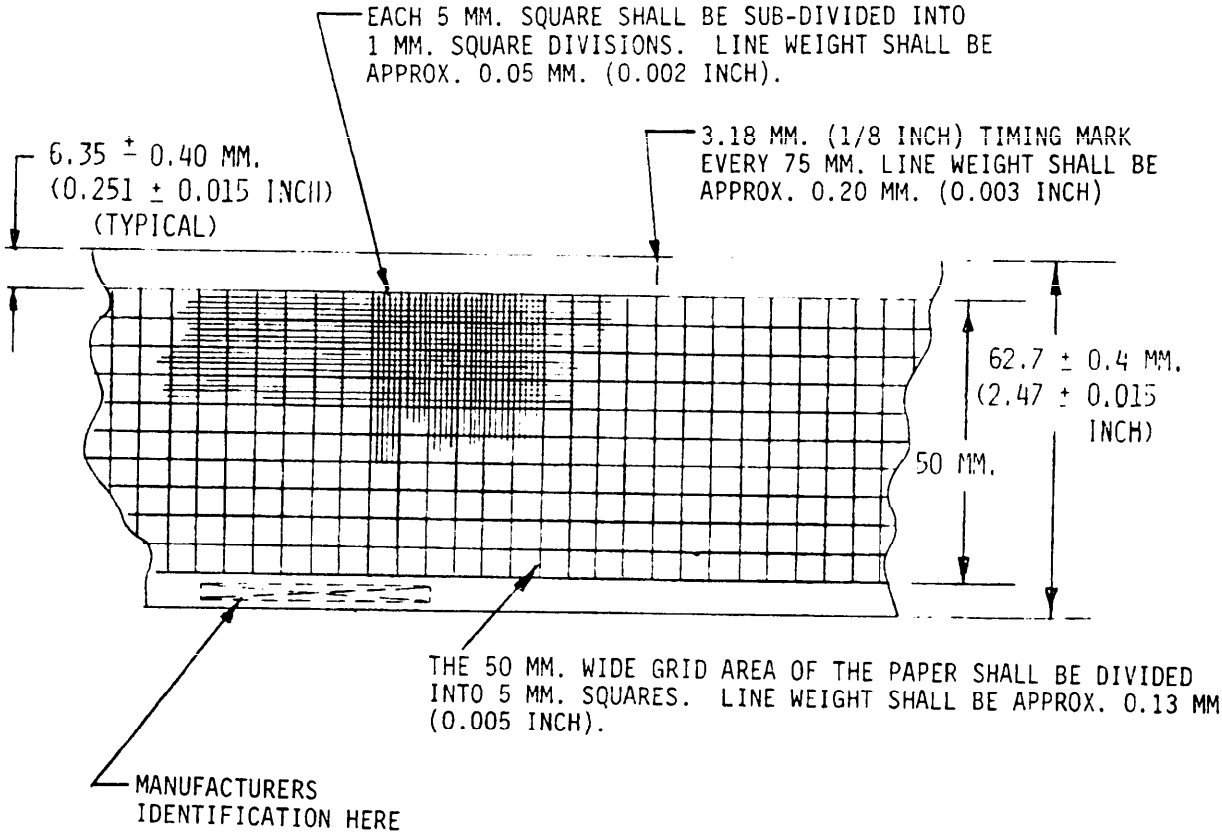
3.5 Identification marking. The paper shall be clearly and legibly marked with the name of the registered trademark of the contractor (see figure 1).

3.6 Workmanship. Workmanship shall be first class throughout. Paper shall be free from defects which detract from its appearance or may impair its serviceability. Paper shall be cut square and edges shall be parallel with the lines printed thereon.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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NOTES

1. ACCURACY OF LINE SPACINGS SHALL BE WITHIN ± 0.3%.
2. AN END OF ROLL MARK SHALL BE LOCATED APPROX. 7.6 METRES (25 FEET) FROM THE END OF THE ROLL. WIND WITH PRINTING TO INSIDE.
3. LENGTH OF PAPER SHALL BE NO LESS THAN 45.1 METRES (148 FEET).

FIGURE 1. Electrocardiograph Recording Paper.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the paper when a first article sample is required (see 3.1). This inspection shall include examination of 4.3. and test of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table II. The unit of product for sampling purposes shall be one roll of paper.

TABLE II. Sampling for examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	S-2	1.0
Total defects (Major and Minor combined)	S-4	4.0
For dimensional examination	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. The acceptance number shall be zero for all tests. The unit of product for sampling purposes shall be one roll of paper.

TABLE III. Sampling for tests.

Characteristic	Requirement	Test Procedure	Inspection level
Thickness	3.2.1	4.4.1	S-1
Basis weight	3.2.1	4.4.2	S-1
Tensile strength	3.2.1	4.4.3	S-1
Water resistance	3.2.1	4.4.4	S-1
Brightness	3.2.1	4.4.5	S-1
High speed writing	3.4.1	4.4.6	S-4
Amplitude doubling	3.4.2	4.4.7	S-4
Buildup of paper coating and degradation of traces	3.4.3	4.4.8	1/

1/ This test shall be performed one time on at least 3 rolls from each lot of finished material.

4.3 Examination. The paper shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table IV. Examination shall not be restricted to the classified possible defects listed below.

TABLE IV. Classification of defects.

Categories	Defects
Major:	
101	Chart spacing incorrect.
102	Specified printing missing, incomplete, illegible.
Minor:	
201	Grid line coloring not as specified.
202	Paper not clean.

4.3.2 Dimensional examination. The paper shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Thickness. The paper shall be tested for thickness in accordance with TAPPI Standard T 411.

4.4.2 Basis weight. The paper shall be tested for basis weight in accordance with TAPPI Standard T 410.

4.4.3 Tensile strength. The paper shall be tested for tensile strength in accordance with TAPPI Standard T 404.

4.4.4 Water resistance. A drop of water shall be applied to the sensitized surface of the paper at room temperature (70° F.). A visual check shall be made for discoloration after the drop of water has evaporated.

4.4.5 Brightness. The unprinted paper shall be tested for brightness in accordance with TAPPI Standard T 452.

4.4.6 High speed writing. The ECG (see 6.5) shall be adjusted in accordance with the manufacturer's instructions. Each roll shall be tested twice. Once with the zero point of the stylus located at the 10 mm amplitude line of the grid and once with the zero point of the stylus located at the 0 mm amplitude line of the grid. Each test shall be 1 minute in duration. The ECG shall be adjusted for a chart speed of 25 mm/sec and a sensitivity of 0 mm/mv. The signal used to excite the input to the ECG shall have a frequency of 50 Hz. and an amplitude of 1 mv.

4.4.7 Amplitude doubling. The ECG (see 6.5) shall be adjusted in accordance with the manufacturer's instructions, the zero-point of the stylus shall be adjusted to coincide with the center-line of the ECG paper, and the chart speed shall be set at 25 mm/sec. With the ECG sensitivity set at mm/mv, the 1mv internal calibration circuit shall be energized for five seconds and the average stylus deflection (in millimetres) shall be recorded. This shall be repeated with the ECG sensitivity set at 10 mm/mv and again at 20 mm/mv. Changing the sensitivity of the ECG from 5 mm/mv to 10 mm/mv and from 10 mm/mv to 20 mm/mv should, in each case, cause a doubling in the average stylus deflection. Maximum deviation from doubling shall be as specified in 3.4.2.

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4.4.8 Buildup of paper coating and degradation of traces. With the ECG (see 6.5) adjusted in accordance with the manufacturer's instructions and the zero point of the stylus adjusted to coincide with the center-line of the ECG paper, the input to the ECG shall be excited with a 0.100 Hz signal. The amplitude of the signal and the sensitivity setting of the ECG shall be adjusted so that the trace extends across the full width of the grid section of the paper. The ECG shall then be operated with no further adjustment at a chart speed setting of 25 mm/sec until 3 full rolls of paper have been used. The traces shall then be examined for the deficiencies noted in 3.4.3.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One box containing two rolls of paper, as specified constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Two rolls of paper shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure of the box shall be as specifie in the appendix of the applicable box specification.

5.1.3 Level C. Two rolls of paper shall be clean, dry and packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Fifty unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Fifty unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged rolls of paper shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

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5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked as specified in MIL-STD-129. The contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The ECG recording paper is intended for use with stock number 6515-00-449-2178, Electrocardiograph during administration of ECG examinations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

6.3 First article. When a first article is required, it shall be tested under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection of the first article.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-584-3076	Paper, Electrocardiograph, Recording, Direct Writing, 150 feet, 2s

6.5 Electrocardiograph machines that have been found to be acceptable are the following:

Burdick Corp., Model EK-SA
Cambridge Instrument Co., Inc., Model VS-4
Hewlett-Packard, Model 1500B
Parke-Davis, Model 2100

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6515-2028

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-37574 PAPER, ELECTROCARDIOGRAPH RECORDING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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